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REVIEW

Performance Enhancing Supplements and Cardiovascular Health

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Abstract

Purpose of Review This review explores the growing use of performance enhancing supplements (PES) with a focus on those used among recreational athletes. We discuss the mechanisms of action, benefits, and cardiovascular risks associated with these substances to inform safe practices and address an important area in sports cardiology.

Recent Findings Anabolic–androgenic steroids (AAS), used to supplement testosterone levels beyond physiological levels, are increasingly popular among recreational athletes, with their misuse linked to serious cardiovascular and metabolic disease, including dyslipidemia, hypertension, arrhythmias, coronary artery disease, and cardiomyopathy. Recent large observational data suggest a more than twofold increase in mortality risk among AAS users. By contrast, randomized clinical data suggest relative cardiovascular safety of testosterone replacement therapy (TRT) in older hypogonadal men, despite a mildly elevated risk of thromboembolic events and arrhythmias. For postmenopausal women, hormonal therapy (HT) may offer improvements in exercise performance and possibly cardiovascular health when initiated early but may increase cardiovascular and thrombotic risk in older women or those with comorbidities. Beyond AAS and hormonal therapies, pre-workout supplements and creatine are widely used among athletes to improve focus, strength, and aid in recovery. Although their excessive use raises concerns, low to moderate consumption of caffeine and creatine appears to carry minimal cardiovascular risk.

Summary PES are increasingly prevalent in sports, offering potential performance benefits but also posing significant cardiovascular risks when misused. Each PES category has unique mechanisms of action and safety profiles. Raising awareness of PES and their cardiovascular risk among athletes and sports cardiologists is critical to mitigate adverse events and maintain athlete health.

Keywords Creatine · Caffeine · Steroids · Anabolic · Athletes · Testosterone

Opinion Statement

The widespread use of PES in both professional and recreational sports pose unique challenges for sports cardiology. The use of PES presents a complex trade-off between potential performance benefits and cardiovascular risks. Each category of PES has a distinct mechanism of action and safety profile. AAS use is associated with a range of adverse cardiovascular effects, including dyslipidemia, hypertension, cardiomyopathy, and thromboembolic events. Individuals using AAS should undergo regular cardiovascular screening, including blood pressure monitoring, lipid profiling, and echocardiographic evaluation. Counseling regarding potential long-term risks is paramount, particularly given emerging evidence linking AAS use to increased mortality risk. Individualized risk assessment is crucial for older individuals using

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TRT or HT. While TRT attenuates age-related decline in cardiorespiratory fitness and appears safe in hypogonadal men with appropriate clinical indications, it should not be prescribed indiscriminately. Similarly, HT in postmenopausal women may help maintain skeletal muscle health and confer cardioprotective benefits when initiated early, however HT can increase cardiovascular risk in older women or those with underlying cardiac comorbidities. HT should be reserved for select individuals with personalized risk assessments. More commonly used PES, like caffeine and pre-workout supplements are generally well tolerated in moderate doses, and may increase endurance performance. However, caffeine poses dose-dependent cardiovascular risks including arrhythmias and hypertension, and excessive intake warrants caution, especially in individuals with genetic predispositions to arrhythmias. Athletes should be educated on safe dosing strategies and potential interactions with other stimulants. Finally, creatine remains one of the most effective and safest PES options for improving strength and power. Current data indicate minimal cardiovascular risk in healthy athletes, though long-term effects require further study. Overall, a risk-stratified approach, emphasizing education and shared decision-making, is essential in managing PES use while prioritizing cardiovascular health (Fig. 1).

Introduction

Performance enhancing supplements (PES) comprise a broad array of substances aimed at improving exercise performance, recovery, and injury prevention. These include Schedule III substances banned under the Controlled Substances Act and World Anti-Doping Agency (WADA) such as anabolic–androgenic steroids (AAS), as well as legal options like creatine and caffeine. The rising trend of PES use among athletes has made this an important, albeit understudied area in sports cardiology. In this review, we discuss PES with a focus on those commonly used among recreational athletes, including AAS, hormone therapy (HT) in older individuals, caffeine/pre-workout supplements, and creatine.

Anabolic–Androgenic Steroids

AAS are synthetic derivatives of testosterone that bind to androgen receptors to induce male sexual characteristics (androgenic properties), and skeletal muscle hypertrophy (anabolic properties). Their misuse—characterized by the use of these substances without a valid medical prescription and at supraphysiologic doses—has become increasingly common in both recreational and professional sports and may be driven by the desire to enhance athletic performance,

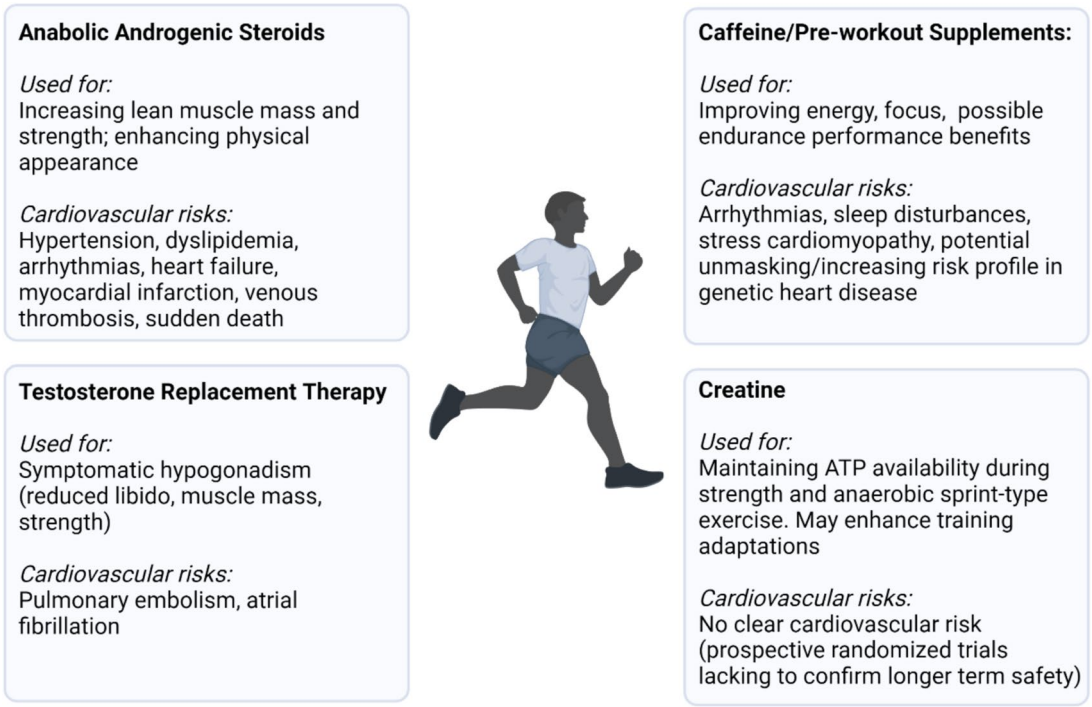

Fig. 1 Summary of commonly used performance enhancing supplements and potential cardiovascular risks. Created in BioRender. Rao, P. (2025) https://BioRender.com/y37a242

build muscle mass, and/or improve physical appearance. A meta-analysis of 187 studies estimated the overall lifetime prevalence of AAS use at 3.3%, with higher rates observed among men (6.4%) and recreational athletes (18.4%) [1, 2]. While underrecognized, AAS use is also prevalent among women. It is reported that up to 4.4% of recreational women athletes and 16.8% of young women bodybuilders use AAS [3].

AAS-induced improvements in strength range from 5–50% of baseline levels, while body weight may rise by 2–5 kg, largely due to gains in lean muscle mass, particularly of type 1 muscle fibers [4]. Although their precise effect on endurance performance is less clear, AAS influences red blood cell production and hemoglobin levels, which in turn may increase cardiorespiratory fitness [5, 6]. However, AAS are associated with a spectrum of adverse cardiovascular and metabolic disease, including hypertension, dyslipidemia, and arrhythmias. Among 100 men (≥18 years) undergoing a cycle of AAS, mean systolic and diastolic blood pressure increased by 6.87 and 3.17 mmHg, respectively, potentially through direct effects on vascular function and sodium retention [6, 7]. In this cohort, LDL cholesterol and ApoB increased 0.45 mmol/L and 18.2 mg/dl respectively, whereas HDL cholesterol and ApoA decreased by 0.40 mmol/L, and 36.6 mg/dl, respectively. Importantly, many of these deleterious risk factor changes were reversible after cessation of AAS, likely due to the restoration of endogenous testosterone levels. Chronic misuse, however, may result in subclinical disease such as left ventricular hypertrophy (LVH) and coronary microvascular dysfunction, as well as adverse cardiovascular events, including myocardial infarction, heart failure, atrial fibrillation, and thromboembolic disease [8–12]. Although subject to confounding, large registry-based Danish study including 1189 suspected AAS users-the majority likely recreational athletes-demonstrated a greater than twofold risk of mortality compared to matched controls (hazard ratio (HR) 2.81; $P<0.001$; mean follow-up: 11yrs) [13]. These data, in addition to well-established AAS effects on multiple other organ systems highlight the importance of appropriate monitoring for individuals that choose to engage in this behavior. Specifically, regular screening of serum hormone levels, hematocrit, and lipid profiles, along with blood pressure and echocardiographic assessments, are important to mitigate the potential risk for cardiometabolic disease.

Testosterone Replacement Therapy (TRT) in Older Men

Though technically classified as an AAS, TRT is intended to restore physiological testosterone levels in hypogonadal older males, distinguishing it from the supraphysiological testosterone levels sought by younger, healthy individuals using AAS. With regards to exercise performance, while AAS/supraphysiological testosterone levels have well-established performance enhancing benefits as outlined above, supplementation of testosterone to maintain physiological levels in older men may only have a modest performance effect. In fact, testosterone supplementation to an exercise regimen did not lead to additional strength and lean muscle mass gains among a cohort of 90 individuals, although TRT attenuated the age-related decline in cardiorespiratory fitness and may be beneficial for individuals who are unable to exercise [14, 15]. Despite its potential benefits on functional capacity, fatigue and exercise intolerance may not always be directly attributed to testosterone deficiency, even in hypogonadal older males. Multiple factors contribute to symptoms of reduced exercise tolerance in this population. TRT should therefore be considered in the context of a comprehensive evaluation, rather than assumed to be a universal solution for exercise intolerance in older males.

Since early observational data demonstrating an inverse relationship between circulating testosterone levels and cardiometabolic disease in older men, testosterone sales increased from 150 million in 2000 to 1.8 billion in 2011 [16]. However, several cardiovascular safety concerns were raised after this uptick in sales, with observed higher rates of cardiovascular disease among older men on TRT [17, 18]. These data led to a more rigorous evaluation of TRT in a randomized clinical trial involving 5246 men aged 45 to 80 years with established or increased risk of cardiovascular disease and hypogonadism [19]. In this study, TRT was noninferior to placebo for the incidence of major adverse cardiac event (testosterone vs placebo: 7 vs 7.3%; P: 0.96). There was, however, an increased risk of pulmonary embolism, acute kidney injury and non-fatal arrhythmias including atrial fibrillation in the TRT group. Of note, the study had a low adherence and retention rate with only 39% completing the full duration of the study drug in each group, therefore biasing towards a null result. Nevertheless, these findings lend support to the cardiovascular safety of TRT in this *specific* population although these data should not be generalized to older men without confirmed hypogonadism.

Hormone Therapy (HT) in Older Women

The menopausal transition is marked by a change in hormone levels, including a decline in estrogen and progesterone, and increase in follicle-stimulating hormone (FSH). HT regimens vary but often supplements estrogen ± progesterone, and less commonly testosterone to mitigate their physiological decline.

From an exercise performance standpoint, estrogen plays a vital role in maintaining skeletal muscle health by promoting repair, and possible mitochondrial function benefits through its effects on muscle satellite cells,

membrane stability, and antioxidant pathways [20]. In the absence of adequate estrogen, postmenopausal women experience greater loss in muscle mass and force-generating capacity compared to age-matched men, highlighting the potential utility of HT in mitigating these declines. Furthermore, low-dose testosterone combined with estrogen therapy in postmenopausal women may enhance cardiovascular health by improving body composition, insulin sensitivity, and lowering LDL cholesterol [21]. Despite these possible benefits, HT has nuanced and complex effects on cardiovascular health. Furthermore, its effect on the cardiovascular system depends heavily on the timing of initiation, formulation, route of administration, and baseline cardiovascular risk of the individual.

Before the landmark Women's Health Initiative (WHI) trial, observational studies showed an association between HT and lower cardiovascular disease risk and mortality. For instance, in the Nurses' Health Study (ages 30–63 years) HT users (estrogen and progesterone or estrogen alone) had a lower incidence of coronary heart disease and cardiovascular mortality compared with never-users [22]. These findings aligned with estrogen's known beneficial effects on the vascular endothelium and lipid modulation [23]. Due to the growing evidence for lower rates of coronary heart disease among women using HT, the WHI was launched in 1992 to specifically address the effects of conjugated equine estrogen ± medroxyprogesterone acetate on cardiovascular disease. This trial, the largest randomized trial of HT to date, yielded surprisingly conflicting results. Individuals who received oral conjugated equine estrogens plus medroxyprogesterone acetate, demonstrated an increased risk of stroke (0.44% per year vs. 0.32% per year with placebo) and pulmonary embolism (0.23% per year vs. 0.15% per year with placebo) [24]. Importantly, the average age of WHI participants was 63 years and many had cardiometabolic comorbidities, raising questions about generalizability to younger, healthier women. Follow-up studies from the WHI and subsequent RCTs highlight the importance of timing in HT initiation. A 2015 Cochrane review confirmed a nearly 50% reduction in coronary heart disease risk in women younger than 60 or within 10 years of menopause who started HT (RR 0.52, 95% CI 0.29–0.96), whereas older women or those initiating HT later showed no benefit, but demonstrated an increased risk of stroke (RR 1.21, 95% CI 1.06 to 1.38) [25]. While further research is needed to identify the exact individuals who may benefit from HT, these findings suggest that HT may be most useful among younger, recently postmenopausal women, however, may increase cardiovascular and thrombotic risk in older women or those with existing comorbidities.

Caffeine, Stimulants, and Pre-workout Supplements

Caffeine (1,3,7-Trimethylxanthine) and other stimulants are among the most used PES, particularly after the removal of caffeine from the WADA list of banned substances in 2004. Between 1987–2004, caffeine was prohibited only in competition, with a urinary concentration threshold of 12 µg/mL set in 1987 to discourage excessive use. However, despite its classification as a banned substance in competition, caffeine use among athletes remained prevalent, ultimately leading WADA to remove caffeine from the prohibited list in 2004 [26]. The Food and Drugs Administration (FDA) recommends a caffeine intake below 400 mg/day, with the average 12-oz cup of coffee containing ~ 120–200 mg of caffeine. While this level of caffeine consumption is nearly ubiquitous in the general population, many pre-workouts formulas often contain greater than 200 mg or combine caffeine with other chemicals with similar stimulant effects (Table 1).

Caffeine is metabolized via the cytochrome P450 pathway. Genetic variations in enzymes involved in this pathway, or the concomitant use of certain drugs affecting this pathway may alter caffeine metabolism leading to inadvertently heightened or prolonged caffeine levels. Common physiological effects, primarily exerted through the blockade of adenosine receptors (subtypes A1 and A2), include elevated heart rate, catecholamine levels, and enhanced beta oxidation indicated by increased circulating free fatty acids [27].

Early seminal work showed that caffeine improved endurance capacity whereby trained cyclists were able to extend their time to exhaustion at ~ 80% of maximal oxygen consumption (VO_2max) from 75 to 96 min after ingesting 330 mg of caffeine [28]. These findings align with more contemporary data, whereby participants consuming doses of 3 to 9 mg/kg demonstrated faster endurance running time trials and greater time to exhaustion compared to placebo [29]. However, these performance enhancing benefits have not been consistently demonstrated and the mechanism by which

Table 1 Common pre-workout supplements and caffeine content

Brand	Caffeine per Serving
5-h energy™	200 mg
Bang Energy™	300 mg
C4™	200 mg
Celsius™	200 mg
G Fuel™	300 mg
Prime™	200 mg
Ghost™	200 mg
Reign™	300 mg
NOS™	160–200 mg
Monster™	160 mg

caffeine may improve exercise capacity remains unclear [27, 30–33].

While low-moderate doses of caffeine may increase performance, excessive caffeine (> 6–9 mg/kg) intake may only have limited additive performance benefit but pose serious cardiovascular risk. Acute consumption can lead to increased heart rate, elevated blood pressure, and arrhythmias due to enhanced sympathetic nervous system activity. Studies often categorize caffeine doses into three groups: low (≤ 3.0 mg/kg), moderate (3.1–6.0 mg/kg), and high (≥ 6.1 mg/kg). Caffeine supplementation increases both the prevalence and magnitude of side effects compared to placebo/control. At low doses, 6–34% experience side effects, most commonly tachycardia and sleep disturbance. At high doses, 8–83% of individuals experience side-effects, in part demonstrating the heterogeneity in response to this supplement [33, 34]. Observational studies have linked high doses of caffeine—levels of which are often found in energy/pre-workout drinks—with rare cases of myocardial infarction, serious arrhythmias, and sudden cardiac death [35]. However, despite caffeine's widespread consumption, albeit in typically low-moderate doses, its perceived association with arrhythmia and coronary heart disease lacks strong evidence. Interestingly, there are emerging data to indicate that excessive caffeine consumption may increase the risk of adverse events in individuals with underlying genetic heart disease, specifically congenital long QT syndrome (LQTS). While the primary results of a small cross-over study including 24 patients with LQTS randomized to consume either two cans of energy drink or control drink did not lead to a significant acute change in QTc in the total cohort, 12.5% demonstrated an increase in QTc ≥ 50 ms after ingestion of the energy drink [36]. Taken together, low-moderate caffeine use may enhance athletic performance without significant long-term harm, while excessive stimulant use should merit caution and screening for underlying comorbidities, presence of genetic heart disease, and symptoms.

Pre-workout supplements often combine caffeine with other compounds such as beta-alanine, taurine, and synephrine, which are marketed to enhance energy, focus, and exercise performance. Each ingredient may exert distinct or synergistic effects with caffeine. For instance, beta-alanine, a nonessential amino acid obtained through diet or synthesized in the liver, has been shown to improve time to exhaustion in trained cyclists exercising at 120% of their VO_2max, though it does not enhance performance in 1 km or 10 km time trials [37]. While data on the cardiovascular risks of beta-alanine are limited, excessive supplementation has been linked to neurotoxicity, myotonia, and respiratory distress, with potential amplification of these risks when combined with other pre-workout components. Historically, few pre-workout ingredients have posed serious health risks. For example, the amphetamine derivative 1,3-dimethylamylamine (DMAA), which was used in popular pre-workout supplement Jack3d™, was associated with severe adverse cardiac and neurological events, including cerebral hemorrhage, stress-induced cardiomyopathy, and sudden death, leading the FDA to remove it from the market in 2012–2013 [38].

Creatine

Creatine is one of the most popular dietary supplements for performance enhancement and recovery [39]. Surveys indicate that up to 40% of young athletes and military personnel use creatine, with its use more common in male strength/power athletes [40]. Approximately half of the daily need for creatine is obtained from the diet (e.g. red meat, seafood), and the rest is synthesized primarily in the liver and kidneys [40, 41]. Creatine is largely stored in skeletal muscle (~95%), with smaller amounts located in the brain and testes (~5%). Within skeletal muscle, approximately one-third of creatine exists as phosphocreatine (PCr), while the remainder is present as free creatine [42]. Creatine plays an important role in maintaining energy availability. Its primary metabolic role is to combine with a phosphoryl group (Pi) to form PCr. The hydrolysis of PCr into Cr + Pi can then be used as a buffer to resynthesize ATP to provide free energy for metabolic activity [43]. This helps maintain ATP availability particularly during maximal effort strength and anaerobic sprint-type exercise.

Through an increase in muscle availability of creatine and PCr, creatine supplementation can improve acute anaerobic bouts of exercise, particularly high-intensity efforts < 30 s duration, and possibly enhance training adaptations [44–47]. In addition to its beneficial effect on exercise performance, long-term creatine supplementation may lead to improvement in muscle strength/power and even aid in recovery from intense exercise. In a cohort of experienced marathon runners, creatine loading prior to a 30 km race attenuated the change in several inflammatory markers, reduced muscle soreness, and abolished the increase in lactate dehydrogenase [48].

Weight gain and fluid retention are common side effects of creatine supplementation, which is important to consider for individuals with pre-existing hypertension, left ventricular dysfunction, or for athletes where elevated body weight may be detrimental for performance. However, to date, there are no convincing data to support concerns regarding a deleterious effect on renal function or cardiovascular outcomes. Indeed, ingesting 10 g/day in healthy athletes did not alter creatine clearance, glomerular filtration rate, tubular resorption, or glomerular membrane permeability in the short to medium term (from 10 months to 5 years) [49]. These findings are consistent with reports demonstrating no significant renal effects in American-Style Football players [47].

Furthermore, meta-analyses confirm its general safety across various populations [50, 51]. Taken together, while creatine supplementation (especially at ~ 5 g/day) appears largely safe and effective in increasing high intensity exercise and strength, its use should be monitored carefully. Further prospective randomized trials are needed to confirm longer term safety profiles, determine effective dosage, and assess for differences in outcomes among men and women.

Conclusion

PESs are increasingly influential in both professional and recreational sports. Each category of PES carries distinct mechanisms of action, performance benefits, and cardiovascular risk. While this article provides a broad discussion of commonly used PES among recreational athletes, we encourage the reader to also review societal position stands and expert consensus statements on the benefits and health risks of these supplements [52, 53]. Increased awareness among athletes, healthcare providers, and sports cardiologists is essential to promote safe practices and address growing public health concerns associated with their misuse.

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  - Findings from this large RCT suggest that testosterone supplementation in men with hypogonadism was noninferior to placebo for the incidence of major adverse cardiac event. There was, however, an increased risk of pulmonary embolism, acute kidney injury and non-fatal arrhythmias including atrial fibrillation in the TRT group.
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  - Systematic review on the prevalence and magnitude of side effects associated with caffeine supplementation in athletes. Findings from this study suggest that using ~ 3.0 mg/kg of caffeine may be the best dosing to maximize benefits and minimize potential side effects.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

Competing Interest The authors declare no competing interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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